GGG-S-69D
13 June 1983
SUPERSEDING
Fed Spec GGG-S-69C
December 29, 1980

FEDERAL SPECIFICATION

SAW, POWER HACK, DRY CUT, PORTABLE

(3-INCH CAPACITY)

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a portable power hack saw, weighing not more than 55 pounds/25 kg, with a bar stock cutting capacity of three inches by three inches and flexible feed control.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CC-M-1807	-	Motor, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)
GGG-B-451	-	Blade, Hand Hack Saw and Blade, Power Hack Saw

Federal Standards:

FED-STD-H28	-	Screw Thread Standards for Federal Services
FED-STD-123	-	Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal Specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

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(Federal Government activities may obtain copies of Federal specification documents and the Index of Federal Specifications, Standards and Commercial Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-T-152	-	Treatment, Moisture and Fungus Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-M-18058	-	Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-I-24092	-	Insulating Varnish, Elèctrical, Impregnating, Solvent Containing.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-889	-	Dissimilar Metals
MS16658	-	Connector, Plug and Duplex Receptacle, Electrical, Power, 2 Pole, 3 Wire, Grounding Type, 15 Ampere, 125 Volts.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF LABOR:

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA):

No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI) STANDARDS:

B11.10	-	Construction, Care and Use of Metal Sawing Machines; Safety Requirements for
Z210.1	-	Metric Practice (ASTM-E-380)

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(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN WELDING SOCIETY (AWS):

AWS - Welding Codes

(Application for copies should be addressed to the American Welding Society, 2501 NW 7th Street, Miami, FL 33125.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC, AGENT:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc, Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one complete saw for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The saw shall be new and of the manufacturer's latest design that shall meet the performance requirements of this specification. The saw shall be a portable, electric motor-driven, self-contained unit. All parts subject to wear, breakage, or replacement shall be accessible for adjustment and repair.

3.2.1 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with means for lubrication. All oil holes and filler plugs shall be readily accessible. The saw shall be furnished with all lubrication quantities or levels filled to the proper capacity. Lubrication reservoirs, where applicable, shall have means for checking levels.

3.2.2 Safety and health requirements. Guards, covers, or other safety devices shall be provided for all parts presenting safety hazards. Guards shall not interfere with the operation of the saw. Guards and covers shall be removable for easy access to guarded parts for necessary servicing and maintenance. Saws shall be constructed in accordance with requirements of the manufacturer's responsibilities section of ANSI B11.10 and the standards promulgated under OSHA applicable to the saws themselves. In cases of conflict between ANSI B11.10 and OSHA Standards, the OSHA standards shall apply. Additional requirements for safety and health shall be as specified (see 6.2 and 6.4).

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3.2.3 Interchangeability. All parts shall be manufactured to such standards as shall permit replacement or adjustment without modification of the saw or any of its parts.

3.2.4 Measurement system. Unless otherwise specified (see 6.2), either the US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2). In this specification, all measurements, dimensions, sizes and capacities are given in the US Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI Z210.1 (ASTM E 380).

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.3.2 Construction. The saw shall be constructed so that the structure shall be capable of withstanding all forces encountered during operation of the saw to its maximum rating and capacity without permanent distortion. The saw shall be complete so that, when installed and connected to the specified source of power, it shall be ready for operation.

3.3.3 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching.

3.3.4 Welding, brazing, and soldering. Welding, brazing and soldering shall be employed only when specified in the original manufacturer's design and shall conform to the standard practices described by the American Welding Society code. Surfaces to be welded, brazed, or soldered shall be clean and free of foreign matter. Welds shall be of recommended proportions and penetration sufficient to maintain design strength of the weld assembly. Welding stresses shall be transmitted without permanent deformation or failure occurring during proof or service loading.

3.3.5 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.6 Surfaces. All surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be clean and free of harmful extraneous materials. External surfaces shall be smooth and all edges shall be free from burrs and sharp edges.

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3.3.7 Painting. Unless otherwise specified (see 6.2), painting and finishing of the saw shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall be clean of all foreign matter detrimental to painting and have at least one coat of primer and one coat of enamel or equivalent applied.

3.3.8 Threads. All threaded parts used on the saw shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3.9 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw-type terminals shall be mechanically secured with a means to prevent loss of tightness.

3.3.10 Gears. All gears shall be constructed from a material suitable for the intended purpose. All gears shall be machined and heat treated by a process that shall impart the hardness and toughness that shall enable the gear train to transmit full-rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.4 Performance and product characteristics.

3.4.1 Performance. With no load, the reciprocating action of the hack saw shall be within the range of 100 to 130 strokes per minute. A stroke is considered as one complete cycle of the saw blade. When loaded, the saw shall cut through a three-inch diameter 1045 annealed steel bar in not more than 36 minutes.

3.4.1.1 Sawing accuracy. The accuracy of the saw shall be determined by the squareness of the cut to the longitudinal axis of a three-inch metal bar or tube. This accuracy shall be within 0.052 inch out-of-square.

3.4.2 Operating positions. The saw shall be a self-supporting unit. It shall operate in any position necessary when clamped to a rigidly supported workpiece by means of its vise or when operated from a table or horizontal surface.

3.4.3 Sawing action. The saw shall draw cut, lifting the saw blade from the work for the return motion. The effective cutting action shall be not less than three inches. The saw shall be capable of cutting steel stock from a square cut and at any angular setting up to and including 45 degrees. Angular cutting shall be accomplished within two degrees of any set-cutting angle.

3.4.3.1 Sawing capacity. The saw shall embody the following sawing capacities:

(a) For square cuts and angles up to and including 22-1/2 degrees, stock with minimum cross-sectional dimensions of three by three inches.
(b) For cuts to angles from 22-1/2 degrees up to and including 30 degrees, stock with minimum cross-sectional dimensions of 2-5/8 by three inches.

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(c) For cuts to angles from 30 degrees up to and including 45 degrees, stock with minimum cross-sectional dimensions of 1-1/2 by three inches.

3.5 Components. The hack saw shall consist essentially of a base, reciprocating saw-blade frame, feed mechanism, electric motor drive, and such other components as are required to perform the cutting operations specified herein.

3.5.1 Base. The base shall be constructed with enough rigidity to support the saw-blade frame, drive motor, and work support vise so that the saw can perform cutoff operations to the accuracies and cutting rates specified herein.

3.5.2 Saw blade frame. The saw-blade frame shall be provided with a means of holding the saw-blade. The depth of the saw-blade frame from the saw teeth shall not be less than 3-1/8 inches. The saw-blade frame shall be capable of adjusting the blade to the proper tension and maintaining the blade in vertical alignment with the vertical traverse. Special tools shall not be required for changing or adjusting the blade. The saw-blade frame shall be provided with a means for locking the frame to the base in order that it can be used as a carrying handle when the saw is not in operation. The saw-blade frame shall reciprocate as required to perform the cutting operations specified herein.

3.5.3 Cutting feed. The saw shall be equipped with a flexible feed (constant-pressure feed) that shall progressively lower the saw-blade through the material stock when cutting. The feed mechanism shall be designed with a feed pressure control selector that shall allow a range of feed pressures to be manually adjusted. The feed-pressure range shall be calibrated to allow cutting of soft thin wall tubing up to and including hard alloys such as solid stainless steel bar stock.

3.5.3.1 Automatic shutoff. Means shall be provided for automatically disengaging the feed mechanism and stopping the reciprocating motion of the saw after the blade has cut through the work. Means shall also be provided for stopping the reciprocating motion at any time during the sawing operation.

3.5.4 Saw-blades. Each saw shall be furnished with three, ten-inch blades, one each of 18, 24, and 32 teeth per inch, in accordance with GGG-B-451, type I, grade A, class 1.

3.5.5 Vise. The vise shall be suitable for clamping either flat or round work of the capacities specified in 3.4.3.1. The vise jaws shall open to a distance of at least 3-1/8 inches. The vise shall be adjustable for holding work at cutting angles from a square cut (90 degrees) to 45 degrees and shall be accurately calibrated within this range. The vise shall be capable of withstanding the loads imposed upon the machine when clamped on the work to be cut.

3.5.6 Electrical requirements.

3.5.6.1 Motor. The motor shall not be less than 1/4 horsepower. It shall operate from a 115 volt, 60 hertz, single-phase AC power source. When specified (see 6.2), the motor shall operate from DC power. The motor shall have

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permanently sealed ball or roller bearings and shall be rated for continuous duty. Unless otherwise specified (see 6.2), the motor shall be an induction squirrel-cage type conforming to CC-M-1807.

3.5.6.2 Switch. An electrical power switch shall be provided for starting and stopping the saw.

3.5.6.3 Power supply cord. Unless otherwise specified (see 6.2), the saw shall be furnished with not less than eight feet of three conductor flexible power cord, No. 18 American Wire Gage (AWG), type SO, SJO, or STO in accordance with NFPA Standard No. 70. This cord shall be provided with a two-pole plug with ground conforming to MS16658.

3.5.6.4 Fungus control. When specified (see 6.2), electrical components and circuit elements shall be treated for fungus resistance in accordance with type II requirements of MIL-T-152, except that the statement of treatment is not required. Windings of the totally enclosed electric motors shall be given two impregnating coats of varnish conforming to the requirements of MIL-I-24092, type M, class 130 through 220 (manufacturer's option).

3.6 Weight. The weight of the complete saw shall not exceed 55 pounds.

3.7 Technical data. Data shall be furnished when specified (see 6.3 and 6.3).

3.8 Workmanship. The saws shall conform to the quality end product specified by the requirements in this specification. In addition, the saws shall be free from rust, cracks, incomplete welds, sharp edges or any other defects that may impair operation, durability, or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item specified in 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.3 and all tests under 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

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4.2.2 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to each production unit offered for acceptance under the contract and shall consist of (a) through (c) as follows. Failure of any unit to pass an examination, test, or inspection shall be cause for rejection of the unit.

(a) Product examination (see 4.3).
(b) Functional test (see 4.4.1).
(c) Packaging inspection (see 4.5).

4.2.2.1 Sampling. When specified (see 6.2), sampling for lot acceptance shall be in accordance with MIL-STD-105 at the Inspection Level and Acceptable Quality Level (AQL) specified in the contract. Samples for the functional test shall be selected at random from the samples that have passed the examination in 4.3.

4.3 Product examination. Each production item or sample shall be visually examined to determine compliance with requirements in Section 3, using the following checklist as a guide:

Examination checklist	Requirement reference
Design features not as specified	3.2 thru 3.2.3
Construction not as specified	3.3.2 thru 3.3.10
Sawing action not as specified	3.4.3
Components not as specified	3.5
Cutting feed not as specified	3.5.3 and 3.5.3.1
Saw blades not as specified	3.5.4
Vise not as specified	3.5.5
Motor not as specified	3.5.6.1
Electrical power switch missing or not as specified	3.5.6.2
Power cord not as specified	3.5.6.3
Weight not as specified	3.6
Technical data not furnished	3.7
Workmanship not as specified	3.8

4.4 Tests.

4.4.1 Functional test. The saw shall be connected to a nominal 115 volt source of power and turned on with no load. All functional parts activated by the electric motor shall be operated to assure proper performance of the saw. Adjustments for feed control and pressure feed shall be made to assure proper function of these features.

4.4.2 Operational test. A 12-inch length of three-inch diameter 1045 annealed steel bar shall be used for this test. The operator shall comply with the handbook of instructions for setting up and operating the saw for a square cut. Not less than three cutoffs shall be made with the same blade to determine the cutoff time. Only the actual cutting time shall be used in determining the cutting rate. The cutting rate shall average not more than 36 minutes (see 3.4.1).

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4.4.3 Accuracy. The last cut in 4.4.2 shall produce the piece to be checked for squareness. The edges shall be removed, not to exceed 1/32-inch radius. The cut surface shall not be out-of-square with the outside diameter in excess of 0.052 inch (see 3.4.1.1). A similar test for squareness of cut shall be made on three inch diameter thin-wall steel tubing. Cutoff time will not be determined, but the accuracy shall not exceed 0.052 inch out-of-squareness. The fine-tooth saw blade shall be used for tube cutting.

4.4.4 Sawing capacity. Steel bars of three different dimensional cross-sections are required for this test. The cross-sections shall be: three by three inches, 2-5/8 by 3 inches, and 1-1/2 by 3 inches (see 3.4.3.1). One square cut and one angular cut shall be made on the three by three inch bar stock and one angular cut made on the remaining bar stock sizes. The length of each bar shall be adequate for at least one cutoff. Angular cuts shall be set at random within the degree range specified for each capacity. Angular sawing shall be accomplished within 2 degrees of the set cutting angle (see 3.4.3).

4.5 Packaging inspection. Each saw shall be examined before and after packaging to determine compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be Level A or commercial, as specified (see 6.2).

5.1.1 Level A. Each saw shall be preserved in accordance with the Level A requirements of MIL-M-18058.

5.1.2 Commercial. Each saw shall be preserved and packaged in accordance with standard commercial practice. The complete package shall be designed to protect the saw against damage during shipment, handling and storage.

5.2 Packing. Packing shall be Level A or commercial as specified (see 6.2).

5.2.1 Level A. Each saw preserved as specified in 5.1 shall be packed in accordance with the Level A packing requirements of MIL-M-18058.

5.2.2 Commercial. Each saw preserved and packaged as specified in 5.1 shall be packed to insure delivery at destination, provide for redistribution by the initial receiving activity, and be acceptable by common carrier under the National Motor Freight Classification and Uniform Freight Classification.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with FED-STD-123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The saw covered by this specification is an electric motor powered, light-duty, self-contained, portable machine intended for use in machine shops, stock rooms, and warehouses to dry-cut bars, rods, pipe, tubing, structural and irregular shapes of ferrous and nonferrous metallic materials.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Special safety devices when required (see 3.2.2 and 6.4).
(d) Measurement system when required (see 3.2.4).
(e) Painting if different (see 3.3.7).
(f) Power source when required (see 3.5.6.1).
(g) Motor if different (see 3.5.6.1).
(h) Power supply cord if different (see 3.5.6.3).
(i) Fungus treatment when required (see 3.5.6.4).
(j) Technical data required (see 3.7 and 6.3).
(k) First article inspection if different (see 4.2.1).
(l) Quality conformance inspection if different (see 4.2.2).
(m) Sampling when required (see 4.2.2.1).
(n) Level of preservation, packaging, packing, and marking required (see 5.1, 5.2, and 5.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on DD Form 1423 incorporated into the contract.

6.4 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

Military Custodians

Army - AL
Air Force - 99

Review Activities

Air Force - 84

User Activities
Army - ME

PREPARING ACTIVITY:

Army - AL

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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